WORKED SOLUTION

STEP II 1997, Question 7

Pure • Curve Sketching

STEP II 1997, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s2-q7

The key idea

The right-hand side depends on x only through x^2, so think of it as a quadratic in $u = x^2$ and complete the square: that gives the largest possible value of y^2 in one line, with no calculus. For the sketch, the equation is unchanged when $x \to -x$ or $y \to -y$, so we only need to understand the first quadrant and then reflect. The places worth pinning down are where the curve meets the axes, the slope at the origin, and the slope where the curve meets the x-axis away from the origin.

Solution

Maximum and minimum values

If $a = 0$ the equation says $y^2 = -x^4$, which forces $x = y = 0$: the "curve" is the single point $(0, 0)$, and the only value of y is 0. From now on take $a \neq 0$. Only a^2 appears, so we may as well think of $a > 0$ and write $|a|$ at the end.

Since $y^2 \geq 0$, we need $x^2(a^2 - x^2) \geq 0$, that is $x = 0$ or $x^2 \leq a^2$. So the curve lives in the strip $-|a| \leq x \leq |a|$. Completing the square in x^2,

$$y^2 = a^2x^2 - x^4 = \frac{a^4}{4} - \left(x^2 - \frac{a^2}{2}\right)^2 \leq \frac{a^4}{4},$$

with equality exactly when $x^2 = \frac{1}{2}a^2$. Hence $|y| \leq \frac{1}{2}a^2$ everywhere on the curve, and both $y = \frac{1}{2}a^2$ and $y = -\frac{1}{2}a^2$ are attained, at $x = \pm|a|/\sqrt{2}$. Therefore

$$\boxed{y_{\max} = \frac{a^2}{2}, \qquad y_{\min} = -\frac{a^2}{2}, \qquad \text{both at } x = \pm\frac{|a|}{\sqrt{2}}.}$$

? **Ask yourself** How does this square with calculus? Where is $\mathrm{d}y/\mathrm{d}x = 0$?

Differentiating implicitly, $2y\dfrac{\mathrm{d}y}{\mathrm{d}x} = 2a^2x - 4x^3 = 2x(a^2 - 2x^2)$. Where $y \neq 0$, the slope is zero exactly when $x^2 = \frac{1}{2}a^2$ (the root $x = 0$ has $y = 0$), and these are the four points found above. The points with $y = 0$ need separate care because we can't divide by y there; we deal with them next. Both methods agree, but completing the square proves at once that the stationary values are the overall maximum and minimum.

Shape of the curve

Take $a > 0$. Solving for y gives two branches,

$$y = x\sqrt{a^2 - x^2} \qquad \text{and} \qquad y = -x\sqrt{a^2 - x^2}, \qquad -a \leq x \leq a,$$

which are reflections of each other in the x-axis. On the first branch,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \sqrt{a^2 - x^2} - \frac{x^2}{\sqrt{a^2 - x^2}} = \frac{a^2 - 2x^2}{\sqrt{a^2 - x^2}}.$$

- **At the origin** the slope is a, and on the other branch it is $-a$. So the curve passes through the origin twice and crosses itself there, with tangents $y = \pm ax$.
- **At** $x = \pm a$ we have $y = 0$, and as $x \to a^-$ the slope tends to $-\infty$. So the curve meets the x-axis at $(\pm a, 0)$ with a *vertical* tangent; it's smooth and rounded there, not pointed.
- **In between**, the first branch rises from $(0, 0)$ to its maximum $\frac{1}{2}a^2$ at $x = a/\sqrt{2}$ and falls back to $(a, 0)$. The other quadrants follow by symmetry in both axes.

The result is a figure of eight lying on its side, with loops in $x \geq 0$ and $x \leq 0$, width $2|a|$ and total height a^2.

A tidy way to see the whole curve at once is to put $x = a\sin t$. Then $a^2 - x^2 = a^2\cos^2 t$, so $y^2 = a^4 \sin^2 t \cos^2 t$, and the curve is traced by

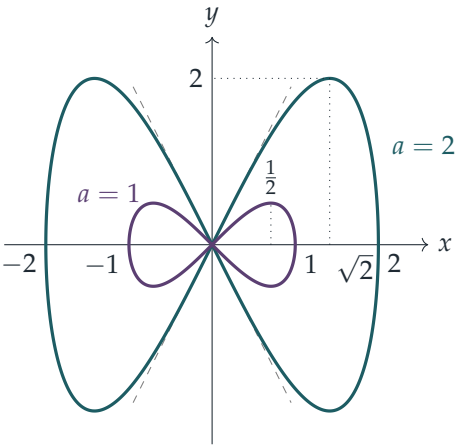
$$x = a\sin t, \qquad y = \tfrac{1}{2}a^2 \sin 2t, \qquad 0 \leq t < 2\pi,$$

which shows again that $|y| \leq \frac{1}{2}a^2$.

The cases $a = 1$ and $a = 2$

	meets x-axis at	tangents at O	extreme points
$a = 1$	$x = 0,\ \pm 1$	$y = \pm x$	$\left(\pm\frac{1}{\sqrt{2}}, \pm\frac{1}{2}\right)$
$a = 2$	$x = 0,\ \pm 2$	$y = \pm 2x$	$\left(\pm\sqrt{2}, \pm 2\right)$

To place the two curves relative to each other: for $0 < |x| \leq 1$ we have $x^2(1 - x^2) < x^2(4 - x^2)$, so the $a = 1$ curve lies strictly inside the $a = 2$ curve, and the only common point is the origin. For example, at $x = 1$ the $a = 1$ curve is at $y = 0$ while the $a = 2$ curve is at $y = \pm\sqrt{3}$.



The dashed lines are the tangents $y = \pm 2x$ for $a = 2$; for $a = 1$ the tangents at the origin are $y = \pm x$, at 45°.

Checks

- A grid search over $|x| \leq |a|$ for 50 random values of a found the largest $|y|$ to be $\frac{1}{2}a^2$ in every case, at $|x| = |a|/\sqrt{2} \approx 0.7071|a|$. For $a = 2$ it gives $y_{\max} = 2.000000$ at $|x| \approx 1.41421$.
- The maximum scales as a^2, as it must: replacing (x, y) by (ax, a^2y) turns the $a = 1$ curve into the general one. So the $a = 2$ curve is the $a = 1$ curve stretched by 2 horizontally and by 4 vertically, and that's why it's so much taller.

Common slips

- Drawing only $y = x\sqrt{a^2 - x^2}$. The equation defines y^2, so both signs of y belong to the graph, giving two loops rather than an S-shape.
- Drawing sharp corners at $(\pm a, 0)$. The slope there is infinite, so the curve crosses the x-axis vertically and smoothly.
- Forgetting that the curve crosses itself at the origin with slopes $\pm a$. These slopes are what tell the two curves apart near O.

Beyond STEP

The origin here is a *node*, a point where an algebraic curve crosses itself: near O the equation looks like $y^2 \approx a^2x^2$, the pair of lines $y = \pm ax$. Classifying singular points by their lowest-order terms like this is the first step in studying algebraic curves. The trick $x = a \sin t$ shows that this curve can be parametrised by trigonometric functions; curves of this kind are called Lissajous figures (here with frequencies in the ratio 1 : 2).

For more practice at sketching curves like this one, and at the STEP way of thinking generally, read Siklos, *Advanced Problems in Mathematics*.

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